Generalized Linear Models

- **Linear regression** is a model that assumes as specific relationships between **input** & **output** - in particular, the assumptions are:
  1. This includes having continuous outputs
  2. The output variable follows a **Normal Distribution with constant variance**
  3. We are predicting the mean μ of the assumed underlying distribution
- What if our output variable is **not continuous** or **normally distributed**?
  - e.g. the number of customers arriving at a store next hour?
  → *Generalized Linear Models extend the framework of LR to handle different types of output variables & underlying distributions*

> **Why would you do this?**
>
> By unifying all sorts of models with one specific formula, you can solve things for this general form and use it for all members of the exponential family

Intuition

When wanting to predict a specific output y, given input features x, we need to make sure to have these assumptions

1. **y | x; θ ~ ExponentialFamily(η):**
   - The underlying assumed distribution for the output variable y, is part of the **exponential family** (e.g. Poission for counts, Bernoulli for binary outcomes)
   - It will be parameterized by the natural parameter η, our model's prediction, that is then used to derive the actual statistical parameters with the link function.

Definition

The general form of a GLM is the following (assuming there is a fixed output variable distribution):

$$p(y;\eta) = b(y)\exp(\eta^T T(y) - a(\eta))$$

This describes the **probability** of observing y, given the natural parameter η - together with other components, that are specific to the chosen exponential family distribution

- $\exp(\eta^T T(y))$ - The core of the model that links η and y for the base probability of y - *positive & high if likely, negative or zero if unlikely*
  - $T(y)$ → If we are predicting multiple statistics, we can have multiple of these
- $b(y)$ - **base measure**, that accounts as a correction factor for the observed value y, but not on η
  - sometimes just 1, depends on distribution
- $a(\eta)$ - **The normalizer/log-partition** function that makes sure that the probability function integrates to 1, making it a valid distribution
- $\eta = \theta^T x = \text{Link}(\text{Model Parameters (like } \mu, or\ \phi))$ - The **natural parameter**
  1. The **familiar linear predictor**, making sure the predictor is linear in its parameters
  2. It is linked to the specific chosen distributions relevant parameters, like ϕ, the probability for $y = 1$, for Bernoulli, or the mean μ for **Gaussian distributions**
  3. As **linear predictions are not bounded**, so η involves a so-called **link function**, that maps these predictions to the specific statistical parameter ranges (e.g. Poisson with $\eta = \log(\lambda) \iff \lambda = e^{\eta} = e^{\theta^T x}$, or the *log-odds* for Bernoulli)

1. Fundamentally it's using the logarithm/exponential to relate b oth and allow for transformation of ranges!

⚠ Usage in real-world

Usually, we don't know the underlying distributions parameters, so what we are trying to estimate are the parameters, given a specific output y and not the other way around.

We then have Likelihood Function and can use MLE to find the best fit (which is basically optimizing the function iteratively via GD, or a closed-form solution)